

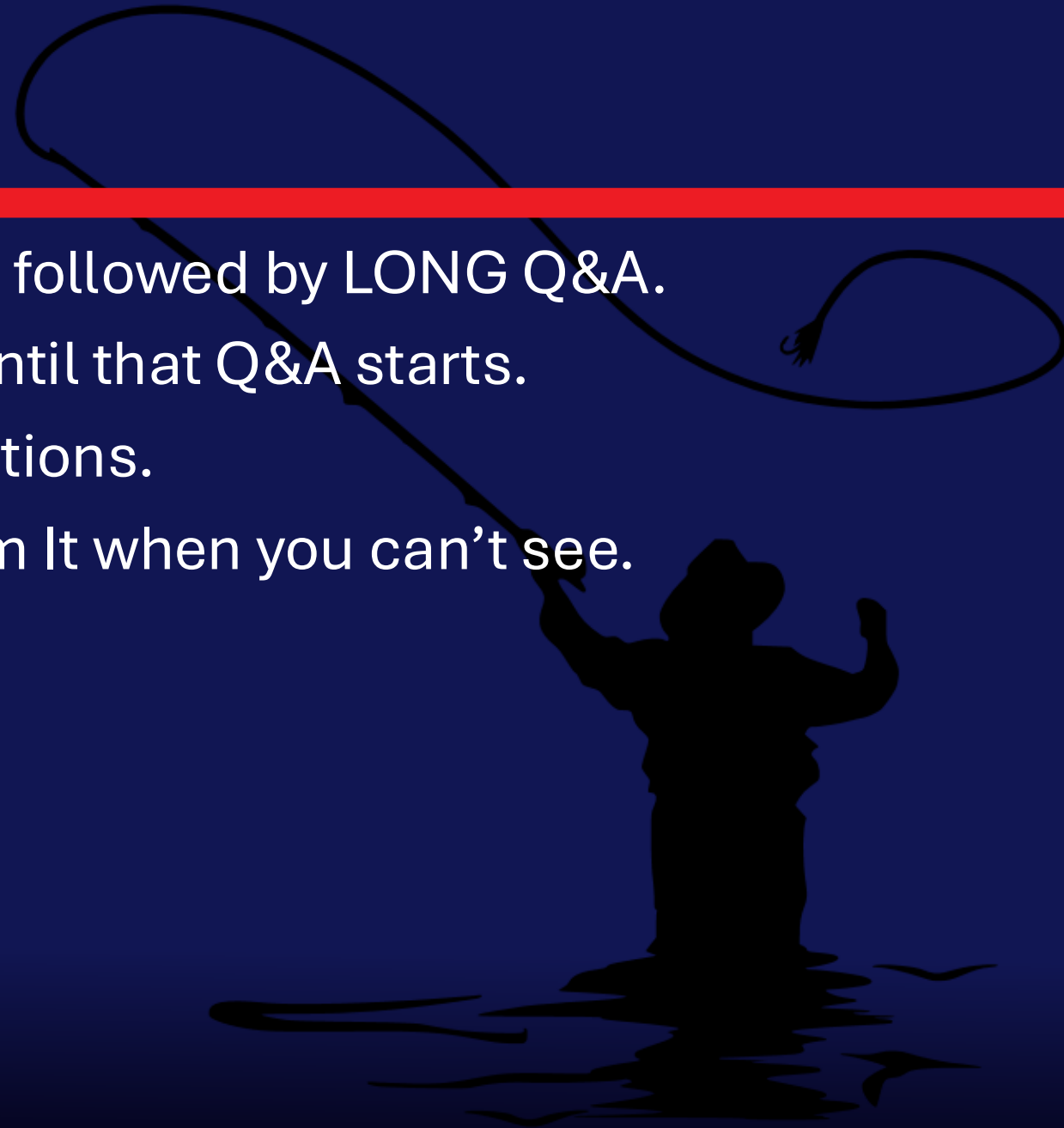
# Mastering App Attach with Azure Virtual Desktop

Leverage App Attach with MSIX packages to efficiently publish and manage applications for Azure Virtual Desktop (AVD) users



# Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A.
- ◆ Please hold detailed questions until that Q&A starts.
- ◆ Feel free to ask clarification questions.
- ◆ Remind the speakers to use Zoom It when you can't see.



# Speakers



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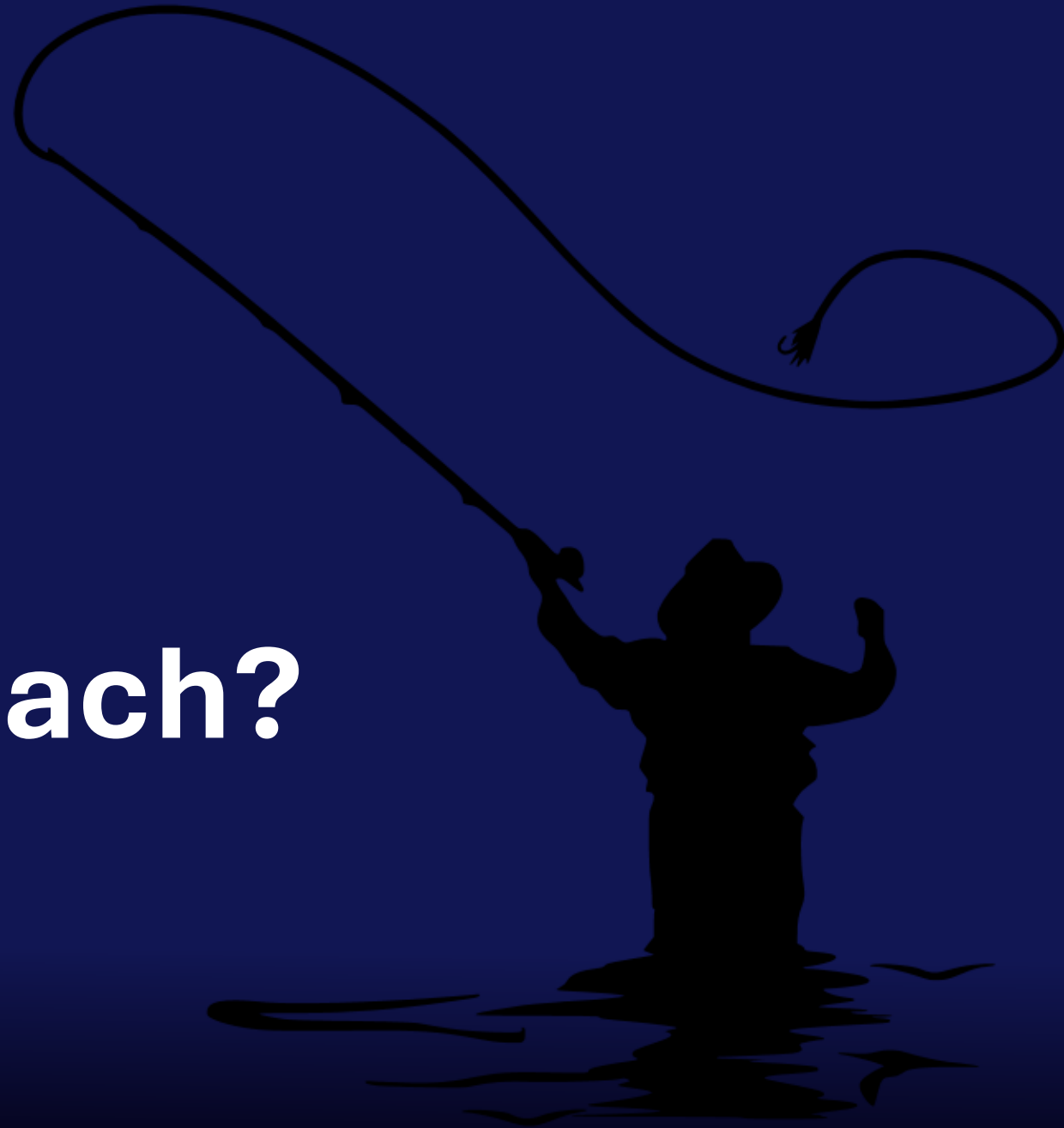


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Can't see our slides? Can't hear? Need to repeat the question? Call us out!

# What is App Attach?

What App Attach is and why it matters.



# What App Attach is...

- ◆ App Attach enables you to dynamically “attach” applications from an application package to a user session in Azure Virtual Desktop.
- ◆ Permissions are applied per application per user, giving you greater control over which applications your users can access during a desktop session. Desktop users only see the App Attach applications assigned to them.
- ◆ Applications can be packaged via **MSIX**, Appx, or App-V.
- ◆ Application packages can be used across host pools.
- ◆ Applications can run on any session host running a Windows OS\* in the same Azure region as the application package.
- ◆ Applications can be upgraded to a new application version with a new disk image without the need for a maintenance window.
- ◆ Users can run multiple versions of the same application concurrently on the same session host.
- ◆ Telemetry for usage and health is available through AVD Insights.

# App Attach and MSIX App Attach

## App Attach

- ◆ Top-level resource in AVD.
- ◆ Windows AD and Entra ID cloud-native identities.
- ◆ Supports MSIX, App-V, and Appx.
- ◆ Run multiple versions of the same application on a session host.
- ◆ Supports the Windows client and Server 2022 and 2025 OS.

## MSIX App Attach

- ◆ Property of an AVD host pool.
- ◆ Windows AD as an identity provider.
- ◆ Supports MSIX.
- ◆ Only one version of the application per session host.
- ◆ Supports the Windows client OS.

# Parts of App Attach

What App Attach is and why it matters.



# MSIX Packages

- ◆ MSI and EXE installs can be packaged in MSIX.
- ◆ Include command line for custom installs.
- ◆ Add application information and details.
- ◆ MSIX packages require a code signing certificate.
  - ◆ Entire certificate chain needs to be trusted on session hosts.
- ◆ MSIX Packaging tool to help build/sign MSIX application packages.





# Signing Certificate

- ◆ All MSIX packages require a valid code signing certificate.
- ◆ Certificates can be obtained from:
  - ◆ A public certificate authority (CA).
  - ◆ An internal enterprise or standalone certificate authority. You need to export the code signing certificate, including its private key.
  - ◆ PowerShell cmdlet New-SelfSignedCertificate that will generate a self-signed certificate. Use for testing/demos.
- ◆ The certificate and certificate change must be trusted on session hosts.
- ◆ Public CA: certificates from a public CA are trusted by default in Windows and Windows Server.
  - ◆ Internal Enterprise CA: AD CS configured as the internal enterprise CA, trusted by default and stored in the configuration naming context of AD DS. When AD CS is configured as a standalone CA, use Group Policy to distribute the root and intermediate certificates to session hosts.
  - ◆ For session hosts joined to Microsoft Entra ID, you can use Microsoft Intune to distribute the root and intermediate certificates to session hosts.
  - ◆ For session hosts using Microsoft Entra hybrid join, you can use either of the previous methods.
  - ◆ Self-signed: install the trusted root to the Trusted Root Certification Authorities store on each session host.

# MSIX Disk Images

- ♦ Image can be VHDX or Composite Image File System (CimFS).
  - ♦ Only use CimFS with Windows 11.
- ♦ MSIX can be converted to MSIX Image CimFS/VHDX using the MSIXMGR Tool
- ♦ Significant performance improvements with CimFS

| Metric               | VHD                      | CimFS        |
|----------------------|--------------------------|--------------|
| Average mount time   | 356 ms                   | 255 ms       |
| Average unmount time | 1615 ms                  | 36 ms        |
| Memory consumption   | 6% (of 8 GB)             | 2% (of 8 GB) |
| CPU (count spike)    | Maxed out multiple times | No effect    |

DEMO

# Create a CIM MSIX Image



# File Share

- ◆ File share can be an Azure Files Storage Account or NetApp Files account.
  - ◆ With NetApp Files, the share and sessions need to be joined to AD DS.
- ◆ For Azure Files and Entra ID joined session hosts:
  - ◆ The **Azure Virtual Desktop** and **Azure Virtual Desktop ARM Provider** service principals need the *Reader and Data Access* Azure RBAC role.
- ◆ For Azure Files with on-premises AD DS Hybrid joined session hosts:
  - ◆ Default share-level “Storage File Data SMB Share Reader” on the storage account.
  - ◆ NTFS permissions applied to allow read access to all session hosts computer objects.
  - ◆ Use the “Domain Computers” group to simplify the deployment.
- ◆ The session hosts must have read access to the file share.
- ◆ Do not use the same storage account used to store FSLogix profile disks.

# Add Image to App Attach

- ◆ To add MSIX images, you need the **Desktop Virtualization Contributor** Azure RBAC role assigned on the resource group as a minimum.
- ◆ Place image in file share, make note of the UNC.
  - ◆ Use the Azure Portal or Azure Storage Explorer to upload the images.



DEMO

# Add MSIX Image to App Attach



DEMO

# App Attach Application in action





# Challenges with App Attach

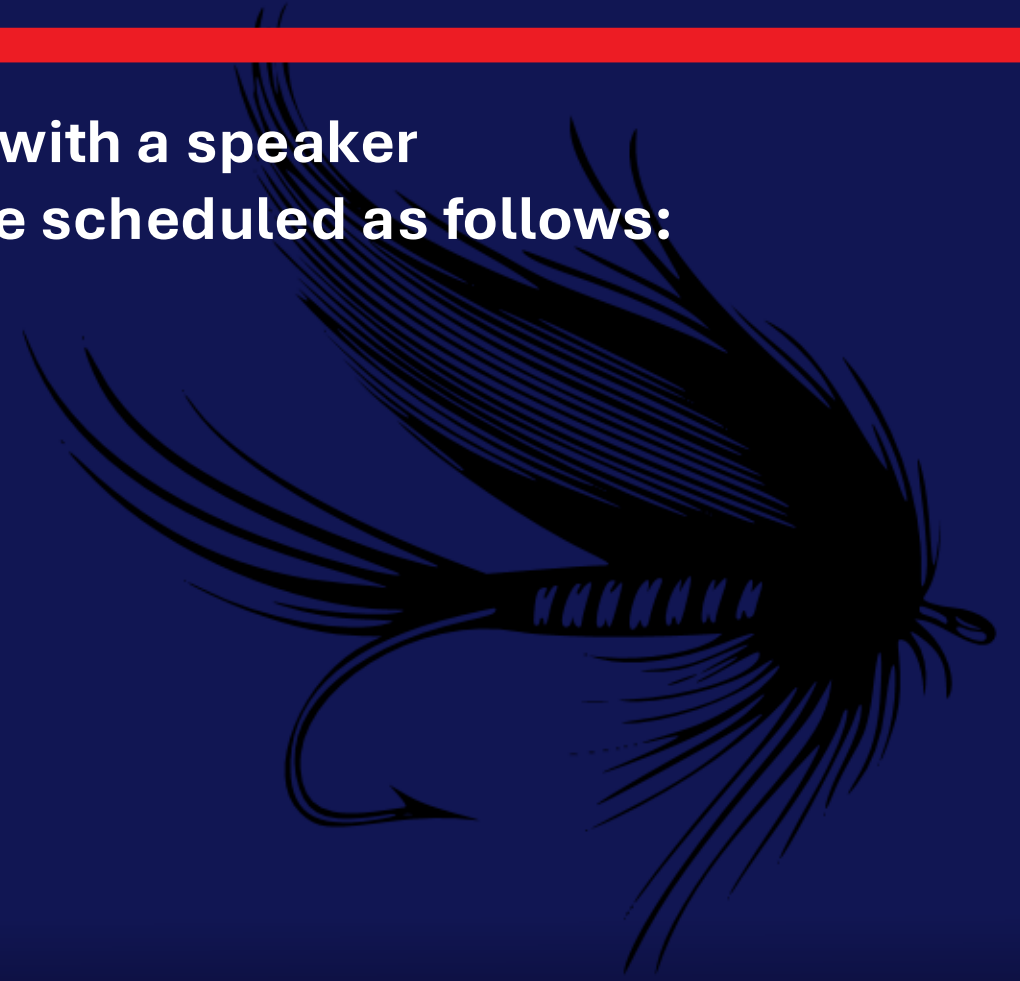
- ◆ Entra Domain Services is not supported with App Attach.
- ◆ Application packaging:
  - ◆ Not all applications can be converted to MSIX.





# Fishing Sessions

Be sure to find Wally and sign up for 1:1 time with a speaker  
If you'd like to "catch" us for a session, we are scheduled as follows:



Don't forget to leave feedback for your speakers!

# SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



# Extended Q&A



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